30 $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ $\mathbf{c} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

(a) Work out as a column vector

(i) $\mathbf{a} + \mathbf{b}$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$

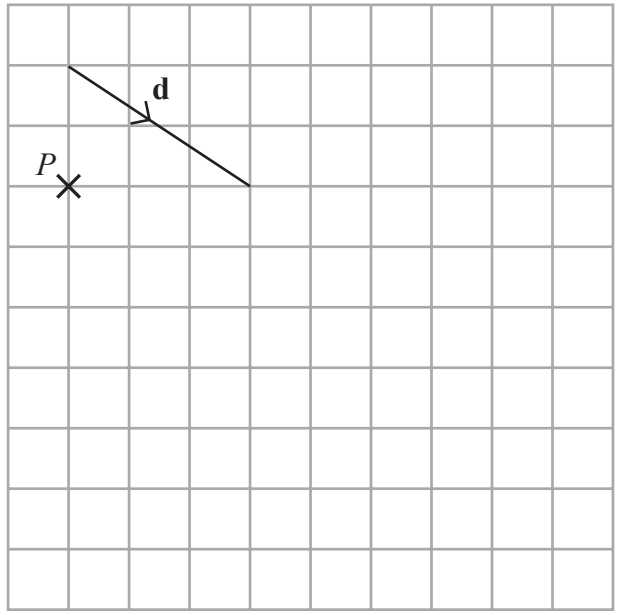
(1)

(ii) $2\mathbf{a} - \mathbf{c}$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$

(2)

The vector **d** is drawn on the grid.



(b) From the point *P*, draw the vector $2\mathbf{d}$

(1)

(Total for Question 30 is 4 marks)